**Problem 9.8.** Equation 9.36 describes the most general **linearly** polarized wave on a string. Linear (or "plane") polarization (so called because the displacement is parallel to a fixed vector $\hat{\mathbf{n}}$) results from the combination of horizontally and vertically polarized waves of the *same phase* (Eq. 9.39). If the two components are of equal amplitude, but *out of phase* by $90°$ (say, $\delta_v = 0$, $\delta_h = 90°$), the result is a *circularly* polarized wave.

(a) At a fixed point $z$, show that the string moves in a circle about the $z$-axis. Does it go *clockwise* or *counterclockwise*, as you look down the axis toward the origin? How would you construct a wave circling the other way? (In optics, the clockwise case is called **right circular polarization**, and the counterclockwise, **left circular polarization**.)[5]

(b) Sketch the string at time $t = 0$.

(c) How would you shake the string in order to produce a circularly polarized wave?

## 9.2 Electromagnetic Waves in Vacuum

### 9.2.1 The Wave Equation for E and B

In regions of space where there is no charge or current, Maxwell's equations read

$$\left.\begin{aligned} &\text{(i)}\ \ \boldsymbol{\nabla}\cdot\mathbf{E}=0, \quad \text{(iii)}\ \ \boldsymbol{\nabla}\times\mathbf{E}=-\frac{\partial\mathbf{B}}{\partial t},\\ &\text{(ii)}\ \ \boldsymbol{\nabla}\cdot\mathbf{B}=0, \quad \text{(iv)}\ \ \boldsymbol{\nabla}\times\mathbf{B}=\mu_0\epsilon_0\frac{\partial\mathbf{E}}{\partial t}. \end{aligned}\right\} \tag{9.40}$$

They constitute a set of coupled, first-order, partial differential equations for $\mathbf{E}$ and $\mathbf{B}$. They can be *de*coupled by applying the curl to (iii) and (iv):

$$\begin{aligned} \boldsymbol{\nabla}\times(\boldsymbol{\nabla}\times\mathbf{E}) &= \boldsymbol{\nabla}(\boldsymbol{\nabla}\cdot\mathbf{E})-\boldsymbol{\nabla}^2\mathbf{E}=\boldsymbol{\nabla}\times\left(-\frac{\partial\mathbf{B}}{\partial t}\right)\\ &= -\frac{\partial}{\partial t}(\boldsymbol{\nabla}\times\mathbf{B})=-\mu_0\epsilon_0\frac{\partial^2\mathbf{E}}{\partial t^2},\\ \boldsymbol{\nabla}\times(\boldsymbol{\nabla}\times\mathbf{B}) &= \boldsymbol{\nabla}(\boldsymbol{\nabla}\cdot\mathbf{B})-\boldsymbol{\nabla}^2\mathbf{B}=\boldsymbol{\nabla}\times\left(\mu_0\epsilon_0\frac{\partial\mathbf{E}}{\partial t}\right)\\ &= \mu_0\epsilon_0\frac{\partial}{\partial t}(\boldsymbol{\nabla}\times\mathbf{E})=-\mu_0\epsilon_0\frac{\partial^2\mathbf{B}}{\partial t^2}, \end{aligned}$$

or, since $\boldsymbol{\nabla}\cdot\mathbf{E}=0$ and $\boldsymbol{\nabla}\cdot\mathbf{B}=0$,

$$\boxed{\nabla^2\mathbf{E}=\mu_0\epsilon_0\frac{\partial^2\mathbf{E}}{\partial t^2}, \quad \nabla^2\mathbf{B}=\mu_0\epsilon_0\frac{\partial^2\mathbf{B}}{\partial t^2}.} \tag{9.41}$$

[5] An elegant notation for circular polarization (or elliptical, if the amplitudes are unequal) is to use a *complex* $\hat{\mathbf{n}}$, but I shall not do so in this book.